



COEP Technological University

(COEP Tech)

A Unitary Public University of Government of Maharashtra

End Semester Examination

Program : S.Y. B.Tech.

Semester: III

Course Code: OEC-24007

Course Name: Linear Algebra

Branch:

Academic Year: 2025-26

Duration: mins

Maximum Marks: 50

Date : 10/11/2025

Student MIS Number :

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Instructions :

1. Write your MIS Number and Branch on the Question Paper.
2. Writing anything else on the question paper is not allowed.
3. Read a given question very carefully and then answer.
4. Mobile phones and calculators are strictly prohibited.
5. Exchange/Sharing of stationery, etc. is not allowed.
6. Figures to the right indicate the course outcomes, full marks and Learning level.
7. Any essential result, formula or theorem assumed for answering questions must be clearly stated.
8. Unless otherwise mentioned, the field $F = \mathbb{R}$ for a given vector space V and 0_V denotes its zero vector.
9. Let $M_{m \times n}(\mathbb{R})$ denote the vector space of matrices of order $m \times n$ with real number entries.
10. Let $\mathcal{B}_1$ be an ordered basis for V and $\mathcal{B}_2$ be an ordered basis for W . Let $T : V \rightarrow W$ be a linear map. Let $[T]_{\mathcal{B}_1}^{\mathcal{B}_2}$ denotes the matrix with respect to $\mathcal{B}_1$ and $\mathcal{B}_2$.

1. Fill in the blanks **without justification**.

- (a) Let $A = \{(1, 0), (1, 1)\}$ and $B = \{(1, -1), (0, 1)\}$. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(a, b) = (2a - b, b)$. Then $[T]_B^A = \dots$ [CO2/3, 1 Mark, L3]
- (b) Let T be a linear map with $T(1, 0) = (1, 4)$, and $T(1, 1) = (2, 5)$. Then $T(2, 3) = \dots$ [CO2/3, 1 Mark, L3]
- (c) Let $A \in M_{2025 \times 2025}(\mathbb{R})$ and O be the zero vector in $M_{2025 \times 2025}(\mathbb{R})$. If A satisfy $A^{2025} = O$ and $A^{2024} \neq O$, then $(I - A)^{-1} = I + \sum_{k=1}^{2024} \alpha_k A^k$. The expression $\sum_{k=1}^{2024} \alpha_k = \dots$ [CO5, 1 Mark, L5]
- (d) Consider the vector space $\mathbb{C}^3$ over $\mathbb{C}$ where $\mathbb{C}$ is the set of all complex numbers and $i = \sqrt{-1}$. Consider the inner product on $\mathbb{C}^3$ to be $\langle (a, b, c), (d, e, f) \rangle = a\bar{d} + b\bar{e} + c\bar{f}$. The value of $\langle (2 + i, i, 1), (-1 + i, 3, -i) \rangle$ is $\dots$ [CO1, 1 Mark, L2]

2. State True or False **without justification**.

- (a) A vector v in a vector space V may have more than one vector w such that $v + w = 0_V$. [CO1, 1 Mark, L1]
- (b) Let λ be an eigenvalue of a linear map $T : V \rightarrow V$. The set $\{x \in V : T(x) = \lambda x\}$ is a subspace of V . [CO1, 1 Mark, L1]
- (c) $(1, 1, 1) \in \text{span}(\{(0, 1, 1), (1, 1, 0)\})$. [CO2, 1 Mark, L2]
- (d) Let $S = \{v_1, \dots, v_n\}$ be an orthogonal set in a vector space V with $\dim(V) = n$. Then $\text{span}(S) = V$. [CO3, 1 Mark, L3]
- (e) Let n be a natural number and $x_1, \dots, x_n \in \mathbb{R}$. Then
$$\frac{|x_1 + x_2 + \dots + x_n|}{\sqrt{n}} \leq \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}.$$
 [CO4, 1 Mark, L4]

3. Let V, W be two finite dimensional vector spaces over $\mathbb{R}$ with dimension m and n . Let $T : V \rightarrow W$ be a linear map, $N(T)$ and $R(T)$ denote its null space and range space. Recall that a function $f : A \rightarrow B$ is one-one if $f(a_1) = f(a_2) \implies a_1 = a_2$. Further a function $f : A \rightarrow B$ is onto if for every $b \in B$ has an $a \in A$ such that $f(a) = b$. In other words $\text{range } f = B$.

- (a) Prove that T is one-one **if and only if** $N(T) = \{0_V\}$. [CO3, 2 Mark, L3]
- (b) Can we remove the finite-dimensional condition on V and W to prove part (a) of this question? [CO4, 1 Mark, L3]
- (c) Prove that T is onto **if and only if** $\dim(R(T)) = n$. [CO2, 2 Marks, L3]
- (d) Let $V = W$. Prove that T is onto **if and only if** T is one-one. [CO5, 3 Mark, L5]

4. This question is concerned with an eigenvalue of a matrix.

- (a) If λ is an eigenvalue of a square matrix A , then prove that λ^3 is an eigenvalue of A^3 . [CO3, 2 Marks, L3]
- (b) Let $A \in M_{3 \times 3}(\mathbb{R})$ have eigenvalues $1, -2, 2$. Find the trace of $A^3 + A^2 + 2A$. (Recall that the trace of a square matrix is the sum of its diagonal entries) [CO3, 3 Marks, L3]

5. This question is concerned with diagonalisability of a matrix. Let n be a natural number.

- (a) Prove that $A \in M_{n \times n}(\mathbb{R})$ is diagonalizable if A has n distinct real eigenvalues. (You can use the fact that the algebraic multiplicity of an eigenvalue λ is always greater than the geometric multiplicity of λ .) [CO3, 2 Marks, L3]
- (b) Diagonalise the matrix $\begin{bmatrix} 0 & -2 \\ 1 & 3 \end{bmatrix}$. Find A^n . [CO4, (5+2) Marks, L4]

6. Let V be an inner product space over $\mathbb{R}$.

- (a) Let $x \in V$. Define norm or length of x denoted by $\|x\|$. [CO1, 1 Marks, L1]
- (b) Prove that $\|x\| = 0$ **if and only if** $x = 0_V$. [CO3, 2 Marks, L3]
- (c) Let us consider the inner product space $V = \{f : [0, 2\pi] \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$ with the usual inner product. Find $\|\sin x\|, \|\sin^2 x\|$. [CO2, 2 Marks, L2]
- (d) Prove that $\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$, for any $x, y \in V$. [CO3, 3 Marks, L3]

7. In each of the inner product spaces with usual inner product, apply the Gram-Schmidt process to set S to get an orthogonal basis for $\text{span}(S)$.

(a) $V = \mathbb{R}^3, S = \{(1, 0, 1), (0, 1, 1), (1, 3, 3)\}$. [CO2, 3 Marks, L2]

(b)

$$V = M_{2 \times 2}(\mathbb{R}), S = \left\{ \begin{bmatrix} 5 & 2 \\ -2 & 0 \end{bmatrix}, \begin{bmatrix} -1 & 2 \\ 3 & 6 \end{bmatrix} \right\}.$$

[CO3, 2 Marks, L3]

8. Let V be an inner product space with dimension n .

(a) Let $\{v_1, \dots, v_n\}$ be an orthonormal set in V . Prove that

$$x = \sum_{i=1}^n \langle x, v_i \rangle v_i.$$

[CO4, 4 Marks, L4]

(b) Let $B = \{v_1, \dots, v_n\}$ be an orthonormal ordered basis for V . Let T be a linear map from V to V and $A = [T]_B^B$. Let A be given by $(a_{i,j})$ where $i = 1, \dots, n$ and $j = 1, \dots, n$. Using 8(a) or otherwise, prove that $a_{i,j} = \langle T(v_j), v_i \rangle$. [CO4, 2 Marks, L5]
